

Eco Clean – Scheduling Solution

Progress Report 3 – Upul Atapattu

1. Summary

Member	Upul Atapattu
Student ID	300392188

Project Implementation Progress

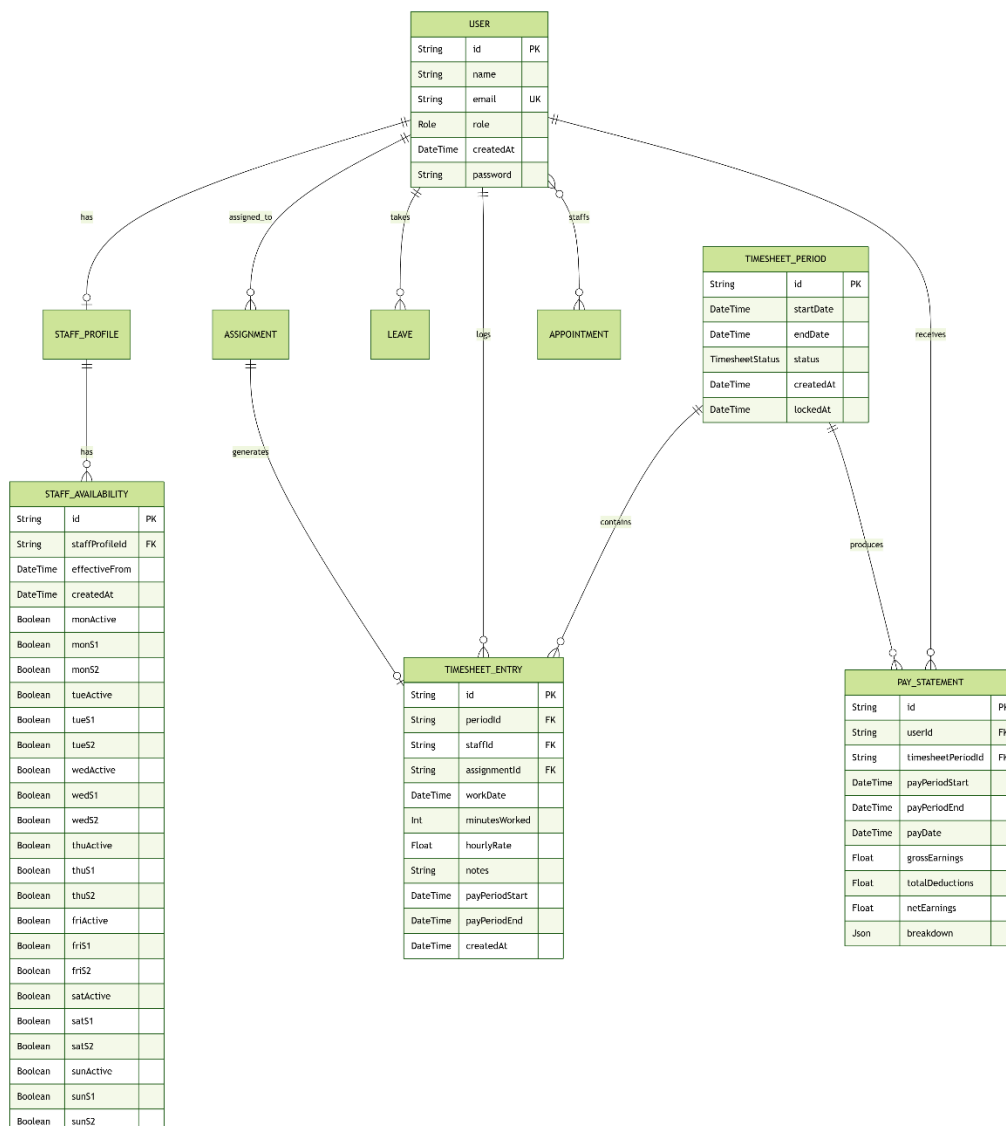
Since the last progress report, the following has been done.

1.1. DB Schema Change

The database schema changes related to staff related interfaces has been changed as follows:

- Staff Availability and Timesheet Period Tables has been modified to match the functional requirements.
- A new table called Pay_statement has been added.

The latest DB Schema changes demonstrated in Mermaid are as follows:



1.2. DB Prisma Migration

After the new database schema, the newly added relationships and entities to the prisma schema design was successfully migrated. We have decided to keep two separate supabase schema, until we fully test the functionality of the new changes brought.

1.3. Payroll and Pay Statement Feature Implementation

To support payroll data management, the admin interface for managing pay periods and payroll entries was designed and developed. This interface allows the administrators at EcoClean to input payroll-related data for each staff member, including regular hours, overtime hours, hourly rates, transport allowances, and multiple deduction categories such as federal tax, employment insurance, Canada Pension Plan (CPP), health deductions, and other adjustments in a single location, which has greater control and visibility.

Once it is entered to the databases, the staff can view their relevant pay stub. This page displays a structured statement of earnings and deductions for each staff member. The layout was designed to resemble a standard payroll statement and includes sections for earnings, deductions, and the final net earnings value.

A key feature which was added to this page is the ability to download the pay stubs as PDF document. The PDF generation functionality allows staff members or administrators to download a formatted pay stub directly from the system.

At the current stage of development, the payroll interface and pay statement components have been implemented on the frontend and structured to integrate with the backend API and database schema. Further integration work is ongoing to fully connect payroll calculations with the timesheet data submitted by staff members.

Following are some of the screenshots of the interfaces.



69 2e Ave N, Roxboro,
Quebec H8Y 2L1, CA

STATEMENT OF EARNINGS AND DEDUCTIONS

Pay Period : 2025-Mar-08 to 2025-Mar-21

Earnings				
	Rate	Units	Amount	Year To Date
Regular	18	80	1540	1540
Overtime	24	10	240	240
Transport Allowance			200	
Gross Earnings			1980	

Deductions				
Federal Tax			-100	
EI			-50	
CPP			-50	
Health			-50	

Net Earnings	1730
--------------	------

Your Availability

Today's Date

March 9, 2026

Employee Name

Effective Date

When does this start?

Weekly Shift Availability *

Day	Available?	Morning (S1)	Afternoon (S2)
Monday	☐	☐	☐
Tuesday	☐	☐	☐
Wednesday	☐	☐	☐
Thursday	☐	☐	☐
Friday	☐	☐	☐
Saturday	☐	☐	☐
Sunday	☐	☐	☐

Comments

Submit

Manage Pay Periods

Period Start Date

15-02-2026

Staff ID	Staff Name	Regular Hours	Regular Rate	Regular Amount	OT Hours	OT Rate	OT Amount	Transport Allowance	Federal Tax	EI	CPP	Health	Other	Total Deductions	Gross Earnings	Net Earnings
STF001	Upul Atapattu	80	18	1440.00	10	24	240.00	200	100	50	50	50	0	250.00	1880.00	1630.00
STF002	Vidarshan	75	20	1500.00	5	30	150.00	100	120	55	60	40	0	275.00	1750.00	1475.00
Totals														3630.00	525.00	3105.00

Generate Pay Statements

1.4. Next Steps

The following are the tasks which are planned by the next progress report deadline (16th March 2026):

- **Finalize Staff Management Components**

Complete the development and testing of staff-related interfaces, corresponding API endpoints, and database CRUD operations, with the objective of stabilizing these features by the next progress report.

- **Begin Implementation of PWA Features**

Start integrating Progressive Web Application (PWA) capabilities for staff-related functions, including features that improve accessibility and usability for staff users when accessing the system through mobile devices.

- **Merge the current development branch with Vidarshan's branch in GitHub**

Resolve any build or TypeScript-related issues that arise during the integration process to ensure a stable and consistent codebase, ready to Vercel Deployment

- **Expanded Test Data Generation**

Populate the database with a larger set of test data using the implemented CRUD operations to thoroughly test system functionality, validate workflows, and identify potential edge cases across the application.

2. Worklogs

Date	Hours	Description
15th January to 2nd March	67.25 hours	Previous Work
5th March 11:00 pm – 12:00 pm	1 hour	Supabase database connection string ran into an error. Had to debug it by resetting credentials and updating the ENV file.
6th March 10:30 pm – 11:30 pm 6:00 pm – 7:00 pm	2 hours	Refactored the StaffAvailability table to align more closely with the intended staff-facing interface to mimic the actual weekly scheduling needs. This has replaced the earlier approach with a consolidated weekly model containing active-day flags and two shift options per day. This is done to reduce complexity in the form submission and to retrieve availability in one row.
7th March 6:00 pm – 7:00 pm 11:00 pm – 12:30 am	2.5 hours	Designed and developed the individual pay stub interface to be linked to the staff profile/pay stubs button. The design was done to resemble a real earnings statement, and the HTML layout was implemented with an option to generate a downloadable PDF version of the pay statement. Tested formatting and downloadable behavior. Modified the prisma scheme to include the data required to generate the paystub by adding PayStatement table.
8th March 9:30 am – 11:00 am 2:00 pm – 3:00 pm 5:30 pm – 6:30 pm	3.5 hours	Designed and implemented the frontend UI for managing pay periods. Built a table-based interface to enter payroll-related values such as regular hours, regular rate, overtime, transport allowance, and deduction items. Added and reviewed the route handlers required for the newly introduced payroll and staff-related entities. Continued backend integration work for pay statements, staff availability, leave, timesheet periods,
9th March 5:00 pm – 6:00 pm 9:00 pm – 10:00 pm	2.0 hours	Worked on debugging some of the UI components related to the payroll. Worked on creating the progress report and updating the AI section and worklogs.
Total	78.25 hours	

3. AI Usage

Please note that only the principal prompt is included in the table. In certain cases, I had put the entire error message or code, in case I was not able to pinpoint the error/issue.

AI	Prompt	Value Addition
Gemini, Pro	<pre>model StaffAvailability { id String @id @default(uuid()) staffProfileId String dayOfWeek Int // 0-6 (Sunday-Saturday) startMinute Int // Minutes from midnight (e.g., 480 for 8:00 AM) endMinute Int // Minutes from midnight effectiveFrom DateTime? effectiveTo DateTime? createdAt DateTime @default(now()) staffProfile StaffProfile @relation(fields: [staffProfileId], references: [id], onDelete: Cascade) }</pre> What would be the implications of changing the schema to <pre>model StaffAvailability { id String @id @default(uuid()) staffProfileId String effectiveFrom DateTime? createdAt DateTime @default(now()) isSunday boolean isMonday boolean isTuesday boolean isWednesday boolean isThursday boolean isFriday boolean isSaturday boolean comments string staffProfile StaffProfile @relation(fields: [staffProfileId], references: [id], onDelete: Cascade) }</pre>	To understand the impact of changing the DB Design for Staff Availability

Gemini, Pro	How to link <pre> export default function EnterAvailabilityPage() { }); to /** * GET: Retrieve all availability rules for a specific staff profile. * Path: /api/staff/[id]/availability */ export async function GET(req: Request, { params }: { params: Promise<{ id: string }> },) { // const session = await getSession(); // // Match security pattern from existing users route // if (!session session.user.role !== 'ADMIN') { // return NextResponse.json({ error: 'Forbidden' }, { status: 403 }); // } try { // UNWRAP the params before using them const { id: staffProfileId } = await params; const availabilities = await prisma.staffAvailability.findMany({ where: { staffProfileId }, orderBy: [{ dayOfWeek: 'asc' }, { startMinute: 'asc' }], }); return NextResponse.json(availabilities); } catch (error) { console.error('GET Availability failed:', error); return NextResponse.json({ error: 'Internal server error' }, </pre>	To Link the API end points with the UI Interfaces
-------------	---	---

	<pre> { status: 500 },); } } } </pre>	
Gemini, Pro	When creating/filling the database PayStatement table from one location to all staff, where should I create my route.ts file for my post method?	To specifically create the API points so the PayStatement table can be updated to all staff at once, giving greater control and visibility
Gemini, Pro	How to just change the width of this table body, so I can try to avoid scrolling. and where should I adjust it? <pre> <Box style={{ width: '100%', maxWidth: 1400, margin: '0 auto' }}> <Table withTableBorder withColumnBorders striped highlightOnHover style={{ minWidth: 1600 }} > <Table.Thead> <Table.Tr bg="green.7"> </Table.Tr> </Table.Thead> <Table.Tbody> {rows.map((row, index) => (<Table.Tr key={row.staffId}> <CellText>{row.staffId}</CellText> <CellText>{row.staffName}</CellText> })} </Table.Tr> </Table.Tbody> </Table> </Box> </pre>	UI interfaces debugging

	<pre> </Table.Td> </Table.Tr> </Table.Tbody> </Table> </Box> </Container>); }</pre>	
Gemini, Pro	What is the meaning of the following error and how to fix it? <pre> <Table.Td>{b.policy}</Table.Td> 244 <Table.Td style={{ textAlign: 'right' }}> > 245 <Text fw={700}>{b.hours.toFixed(2)} hours</Text> ^ 246 </Table.Td> 247 </Table.Tr> 248)})}</pre>	UI interfaces debugging
Gemini, Pro	Why can't I change the date. What should I do to get a date picker. Just point to the place I should make the changes	UI interfaces debugging
Gemini, Pro	How to make sure that we can only choose future dates?	UI control improvement
Gemini, Pro	<pre> let leaveTypeEnum = "PERSONAL"; if (reason.includes("SICK")) leaveTypeEnum = "SICK"; if (reason === "VACATION") leaveTypeEnum = "VACATION"; Do we have to update the prisma ENUMs to avoid this IF?</pre>	Debugging
Gemini, Pro	I want to display my leave in the fullcalender part of the UI. Therefore, I believe I should create a GET method to retrieve data in	UI control improvement

	my route.ts file. After that, I believe it should be adjusted in my page.tsx in the page.tsx, I need to update this in the leave remaining in the balance card too	
Gemini, Pro	What is the meaning of Object literal may only specify known properties, and 'monActive' does not exist in type 'Without<StaffAvailabilityCreateInput, StaffAvailabilityUncheckedCreateInput> & StaffAvailabilityUncheckedCreateInput'.ts(2353) index.d.ts(19710, 5): The expected type comes from property 'data' which is declared here on type '{ select?: StaffAvailabilitySelect<DefaultArgs> null undefined; omit?: StaffAvailabilityOmit<DefaultArgs> null undefined; include?: StaffAvailabilityInclude<...> ... 1 more ... undefined; data: (Without<...> & StaffAvailabilityUncheckedCreateInput) (Without<...> & StaffAvailabilityCreateInput); }'?	UI interfaces debugging
Gemini, Pro	I'm getting the following error, What does it means and how to fix this? <pre> <Box> 124 <Text fw={600} mb={10}>Weekly Shift Availability *</Text> > 125 <Card withBorder radius="md" p={0}> ^ 126 <Table verticalSpacing="sm" horizontalSpacing="md"> 127 <Table.Thead> 128 <Table.Tr bg="gray.0"> </pre>	UI interfaces debugging
Gemini, Pro	How to resolve the followign error?Argument `dayOfWeek` is missing. <pre> at <unknown> (app\api\staff\[id]\availability\route.ts:52:60) at async POST (app\api\staff\[id]\availability\route.ts:52:29) 50 } 51 > 52 const newAvailability = await prisma.staffAvailability.create({ </pre>	API endpoint debugging for Staff Availability

	53 data: { 54 staffProfileId: staffProfileId, 55 effectiveFrom: new Date(body.effectiveFrom), { clientVersion: '7.4.1' }	
Gemini, Pro	Object literal may only specify known properties, and 'monActive' does not exist in type 'Without<StaffAvailabilityCreateInput, StaffAvailabilityUncheckedCreateInput> & StaffAvailabilityUncheckedCreateInput'.ts(2353) index.d.ts(19710, 5): The expected type comes from property 'data' which is declared here on type '{ select?: StaffAvailabilitySelect<DefaultArgs> null undefined; omit?: StaffAvailabilityOmit<DefaultArgs> null undefined; include?: StaffAvailabilityInclude<...> ... 1 more ... undefined; data: (Without<...> & StaffAvailabilityUncheckedCreateInput) (Without<...> & StaffAvailabilityCreateInput); }' 	Debugging StaffAvailability DB Design Changes
Gemini, Pro	how to restart typescript in vscode? can we keep the terminals as it is?	VS Code TS Server Restart, without relaunching everything
Gemini, Pro	Can we simplyfy the payStatement Table? <pre> model PayStatement { id String @id @default(uuid()) userId String timesheetPeriodId String payPeriodStart DateTime payPeriodEnd DateTime grossEarnings Float totalDeductions Float netEarnings Float companyName String? companyAddress String? createdAt DateTime @default(now()) updatedAt DateTime @updatedAt user User @relation(fields: [userId], references: [id], onDelete: Cascade) timesheetPeriod TimesheetPeriod @relation(fields: </pre>	To Improve the PayStatement Table

	<pre> [timesheetPeriodId], references: [id], onDelete: Cascade) lines PayStatementLine[] @@index([userId]) @@index([timesheetPeriodId]) } model PayStatementLine { id String @id @default(uuid()) payStatementId String category PayLineCategory label String rate Float? units Float? amount Float yearToDate Float? payStatement PayStatement @relation(fields: [payStatementId], references: [id], onDelete: Cascade) @@index([payStatementId]) } I only need one. Don't need CompanyName, Address as they are fixed Don't need the updated at too </pre>	
Gemini, Pro	I'm going to put payPeriodStart DateTime payPeriodEnd DateTime in my timesheetEntry. So I need to get it from TimesheetEntry to my paystatemetn. How to write the relationship. Play, payPeriod supposed to be 5+date from payPeriodEnd	To Modify the PayStatement Table
Gemini, Pro	Employ name to be picked up based on the UserID, which is to be taken from the auth session. How do add this to this code?	To Understand how to get the UserID from a auth session

Gemini, Pro	I'm getting the following error. Lets fix that <pre><Box> 124 <Text fw={600} mb={10}>Weekly Shift Availability *</Text> > 125 <Card withBorder radius="md" p={0}> ^ 126 <Table verticalSpacing="sm" horizontalSpacing="md"> 127 <Table.Thead></pre>	Debugging UI Interfaces
-------------	--	---------------------------------------

4. References

- BullMQ - Background Jobs processing and message queue for NodeJS. (2018). Bullmq.io. <https://bullmq.io/>
- Expo. (n.d.). Expo. Expo. <https://expo.dev/>
- Home - Nettoyage Eco Vert. (2026, January 3). Nettoyage Eco Vert. <https://nettoyageecovert.ca/>
- Mantine. (2025, May 5). *Version 8.0.0* (Changelog). Mantine.dev. <https://mantine.dev/changelog/8-0-0/>
- Prisma. (n.d.). *Introduction to Prisma*. Prisma Documentation. <https://www.prisma.io/docs>
- Redis. (n.d.). Redis. Redis.io. <https://redis.io/>
- Vercel. (2026, February 20). *Getting started (App Router)*. Next.js Documentation. <https://nextjs.org/docs/app/getting-started>
- Supabase. (n.d.). *Supabase – The open-source Firebase alternative*. <https://supabase.com/>